Meaning less qty

- quantitative data → Numeric
- Qualitative data → Non numeric
- chronolgocial → Time

variable

- qualitative (Non numeric) or categorical
  - ex :- colours, yes/no, Image
- quantitative (Numeric)
  - Integers
  - Real No.

scales of measurment — virtual scale.

Properties.
- • identification
- • magnitude
- • Equal Intervals
- • True zero (min value)

1) Nominal scale
   • identifity.
2) ordinal scale :- • identification, magnitude. • order wise, ascendin or descending

} qualitative data

3) Interval scale :- it don't have ~~no zero~~ No true zero → both qualitative & quantitative.

4) Ratio scale :- it posses true zero & all propertys → ~~quanti~~ quantitative

theory

How many group?
bin size = ?
n = no. of obs n.
k = no. of bins
w = bin size
$w = \frac{Range}{k}$
$= \frac{max - min}{k}$

Real nos, decimal
discrete
↓
intervals
groups/classes/bin

Optimize k
sturge's formula
$\boxed{\therefore k = 1 + 3.322 k \log_{10} n}$

min = 10
max = 66
width = 5

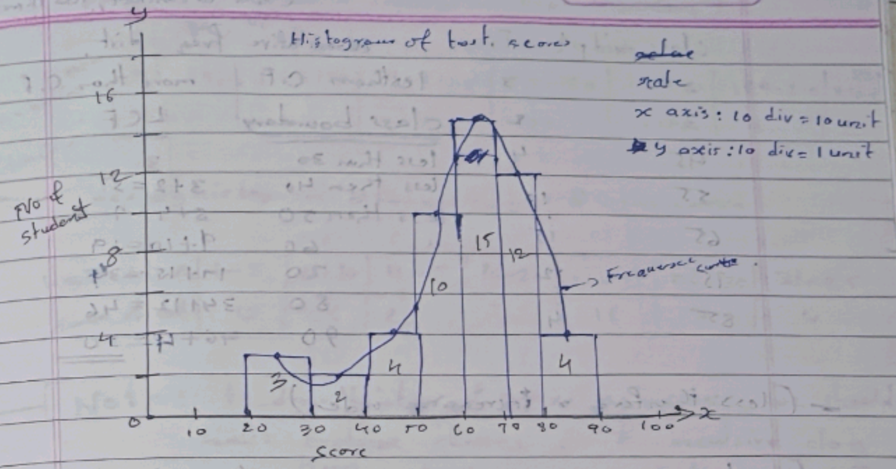
classes		classes	
10 - 15		10 - 14	
15 - 20		15 - 19	
20 - 25	→ Exclusive class	20 - 24	→ inclusive class
25 - 30		25 - 29	
30 - 35		30 - 34	
		35 - 39	
:		:	
65 - 70		65 - 69	
LL lower limit, UL upper limit		LL Lower lim, UL upper limit	

1) List & describe different types of vanbles used in collecting data
2) list different ~~scales~~ scales of measuring data

L.P 3)

~~scores~~	20 - 30	30 - 40	40 - 50	50 - 60	60 - 70	70 - 80	80 - 90
no ST	3	2	4	10	15	12	4

Histogram :- to the histogram we need exclusive classes
- consider
- frequence along y axis
- draw column or a bar for each class whose highth is same it class frequency
- width of each column bar, or regctangle is same as class width.



~~scale~~
scale
x axis : 10 div = 10 unit
y axis : 10 div = 1 unit

Draw frequence curve
plot class frequen against → class midpt $= \frac{L.L + U.L}{2}$
class mid point

⇐ draw relative freque histogram / relative freque curve

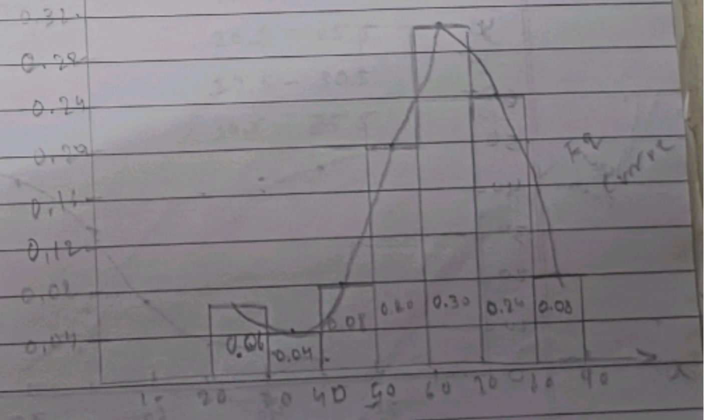
class	F	
20 - 30	3	0.06
30 - 40	2	0.04
40 - 50	4	0.08
50 - 60	10	0.20
60 - 70	15	0.30
70 - 80	12	0.24
80 - 90	4	0.08
	40	1

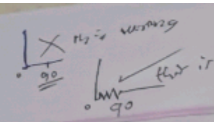
$N = 50$

$Rel\ freq = \frac{freq}{N}$

Realtive freq $= \frac{freq}{N}$

Imp





:- A kink or a zig-zag curve

realitive freq: if we have unequal width then we use realitive Frequence curve.

Numerical discriptions :-

single not summary

Pre-requisite for an ideal measure

- Rigid defination
- easy to understand.
- easy to calcuate.
- Based on all the observation
- suitable for further mathamatical treatment
- least affected by sampling fluctuations & extreme valuez

central value → Measure of central Tendency

spreadness → Measure of Dispresion

Measure of central tendancy

1) Arithmetical Mean (mean or Average)

$x_1, x_2, \ldots x_n \rightarrow n$ order

$$\bar{x} = \frac{x_1 + x_2 + \ldots x_n}{n}$$

$$= \frac{\sum_{i=1}^{n} x_i}{n}$$

for grouped date

$$\bar{x} = \frac{\sum f_i x_i}{N}$$

Where $N = \sum f_i$

2) Median :- middle most value of data when arranged in order

(i) arranged data in ascending order

(ii) Median = value of $\left(\frac{n+1}{2}\right)^{th}$ observation

For Grouped data

$$\text{Median} = L + \left[\left(\frac{N/2 - m}{f}\right) \times h\right]$$

L = lower limit of midean class

m = less then cumulative freq just prisiding of class

H = width of midean class

3) Mode : Most repeated value

Merely by inspection

for Grouped data

$$\text{mode} = L + \left[\left(\frac{f_1 - f_0}{2f_1 - f_0 - f_2}\right) \times h\right]$$

L Lower Limit of model class

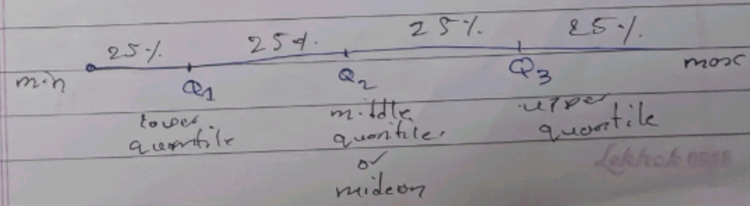
f_1 = freqcee of the model class

f_0 = freque just prisiding model class

f_2 = freq just susceding mode class

H = width of model class

- Positional Averages :-
  - Quartiles :- Divide data into 4 equal parts



Q_i = value of $i \times \left(\frac{n+1}{4}\right)^{th}$ observation.

$i = 1, 2, 3$

For grouped data

$$Q_i = L_i + \frac{\frac{i \times N}{4} - m_i}{f_i} \times h_i$$

$i = 1, 2, 3$

- Measure of Dispersion :-
  - Range = Max − min

Imp
- variance : $(\sigma^2) = \frac{\sum (x_i - \bar{x})^2}{n}$

$$\sigma^2 = \frac{\sum (x_i - \bar{x})^2}{n}$$

$$= \frac{\sum x_i^2}{n} - (\bar{x})^2$$

$$= \frac{x_1^2 + x_2^2 + \dots + x_n^2}{n} - (\bar{x})^2$$

Standard deviation : (σ)

$$\sigma = \sqrt{\text{Variance}}$$

- coefficient of variation $(C.V) = \frac{S.D}{Mean} = \frac{\sigma}{\bar{x}}$

1) calculate coeff

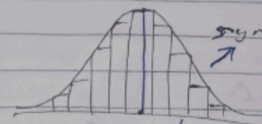
x (i) 4, 5, 6, 6, 6, 7, 8, 9, 10, 12, 15

y (ii) 2, 3, 5, 6, 7, 8, 9, 20, 30, 40, 50

z (iii) 3, 5, 6, 6, 7, 8, 9, 10, 10, 11, 12, 15, 15, 18.

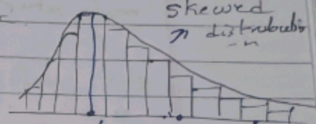
- Skewness

symmetry



symmetric

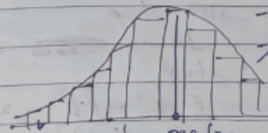
mean = med = mode



+ive positively skewed distribution

mode mid mian

mean > med > mode



−tive negatively skewed distribution

mean mid mode

mean < mid < mode

(Graph is abstract in nature) → disadvantage

coefficient of skewness

Karl pearson method

$$S_K = \frac{Mean - mode}{S.D} \qquad -3 \le S_K \le 3$$

S.D standard Deviation

Bowley

$$S_B = \frac{Q_3 + Q_1 - 2Q_2}{Q_3 - Q_1} \qquad -1 \le S_B \le 1$$

mode = most repeated value.

For nominal → mode — mode is also possible

For ordinal → midian — mode & midian also possible.

ratio → mean

1) i) 4, 5, 6, 6, 6, 7, 8, 9, 10, 12, 15

x :-

ii) mean = $\bar{x} = \frac{\sum x_i}{n}$